

Day 5 — The 12 Rāśis

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will recite the 12 *rāśi* names in order in Sanskrit, match each to its Greek/English equivalent, and identify the approximate calendar month during which the Sun is in each *rāśi* (using the *nirayana* / sidereal system).

Materials

- A large printed circle divided into 12 equal segments (the *rāśi*-wheel template)
- Coloured pencils / sketch pens
- A printed reference table of the 12 *rāśi* names with English translations (from the glossary)
- Student notebooks

Vocabulary introduced today

- *rāśi* (IAST: *rāśi*; lit. “heap, mass; a sign of the zodiac”) — one of 12 thirty-degree-wide segments of the **ecliptic** (the Sun’s annual path on the sky). The Sun is in each *rāśi* for about 30 days.

Lesson flow

Warm-up (5 min)

- Prompt: “What’s your ‘star sign’ according to newspaper horoscopes?” — students call out.
- Teacher: “Those English names — Aries, Taurus, Gemini — they have Sanskrit equivalents. The Indian tradition has 12 *rāśis*, and the Sun visits one each month. Today we learn them.”

Core (20 min)

1. **The ecliptic is divided into 12 equal segments.** Each segment is $360^\circ / 12 = 30^\circ$ of the sky. The Sun crosses each in about 30 days (well, 30.44 on average — see Day 6 footnote).
2. **The 12 *rāśi* names in order.** Write on the board:

#	Sanskrit (IAST)	Greek/Latin/ English	What the symbol depicts
1	<i>Meṣa</i>	Aries	Ram
2	<i>Vṛṣabha</i>	Taurus	Bull
3	<i>Mithuna</i>	Gemini	Twins

#	Sanskrit (IAST)	Greek/Latin/ English	What the symbol depicts
4	<i>Karka / Karkaṭa</i>	Cancer	Crab
5	<i>Siṃha</i>	Leo	Lion
6	<i>Kanyā</i>	Virgo	Maiden
7	<i>Tulā</i>	Libra	Scales
8	<i>Vṛścika</i>	Scorpio	Scorpion
9	<i>Dhanus</i>	Sagittarius	Bow / Archer
10	<i>Makara</i>	Capricorn	Sea-goat
11	<i>Kumbha</i>	Aquarius	Water-pot
12	<i>Mīna</i>	Pisces	Fishes

Notice: the Sanskrit and Greek images are *nearly identical*. The bull, twins, crab, lion, scales, scorpion, archer, fishes — same animals. This is not a coincidence; the two systems were in contact during the Hellenistic period (last few centuries BCE) and there was real exchange. Day 9 will compare them in detail.

3. **When is the Sun in each rāśi?** Roughly (in the *nirayana* / sidereal system used by traditional Indian almanacs):

<i>Rāśi</i>	Sun is here approximately
<i>Meṣa</i>	14 Apr → 14 May
<i>Vṛṣabha</i>	14 May → 14 Jun
<i>Mithuna</i>	14 Jun → 16 Jul
<i>Karka</i>	16 Jul → 16 Aug
<i>Siṃha</i>	16 Aug → 16 Sep
<i>Kanyā</i>	16 Sep → 17 Oct
<i>Tulā</i>	17 Oct → 16 Nov
<i>Vṛścika</i>	16 Nov → 15 Dec
<i>Dhanus</i>	15 Dec → 14 Jan
<i>Makara</i>	14 Jan → 13 Feb
<i>Kumbha</i>	13 Feb → 14 Mar
<i>Mīna</i>	14 Mar → 14 Apr

Each crossing is a *saṁkrānti* (Day 6).

4. **Paraphrase from the Sūrya-siddhānta** (re-anchoring the citation):

The Indian astronomical text **Sūrya-siddhānta** divides the ecliptic into 12 equal *rāśis* of 30° each and gives the Sun's daily angular speed — about 0°59'8" per day on average — which corresponds to one full circuit in ~365.25 days. — paraphrased from Sūrya-siddhānta, chapter on the Sun's motion (see `sources.md`)

Activity (15 min) — Begin the *rāśi*-wheel + formative quiz Part B

Part 1 (10 min) — start the wheel:

- Distribute the *rāśi*-wheel template (a circle divided into 12 wedges, each labelled with a number).
- Students write the Sanskrit name in each wedge in clockwise order (starting from *Meṣa* at the top).
- Below the Sanskrit name, the English / Greek name.
- Below that, the calendar month-range (e.g. *Meṣa* — 14 Apr → 14 May).
- They will finish coloring and decorating during Days 8–9 project work.

Part 2 (5 min) — Formative Quiz Part B: see quizzes/formative.md. Collect.

Wrap-up (5 min)

- Choral recitation: students read the 12 names aloud together, twice.
- Preview Day 6: “Tomorrow — *saṁkrānti*. What the Sun does on the day it crosses from one *rāśi* to the next. There are ~12 *saṁkrāntis* per year. The most famous one is **Makara Saṁkrānti** — around 14 January.”

Student-facing content

The ecliptic is divided into 12 equal segments — the *rāśis*.

Each *rāśi* is 30° of sky. The Sun spends ~30 days in each, and ~365 days to complete the cycle.

The 12 names, in order, are:

Meṣa, Vṛṣabha, Mithuna, Karka, Siṁha, Kanyā, Tulā, Vṛścika, Dhanus, Makara, Kumbha, Mīna.

They map directly onto the Greek/English: Aries, Taurus, Gemini, Cancer, Leo, Virgo, Libra, Scorpio, Sagittarius, Capricorn, Aquarius, Pisces.

Homework

- Memorise the 12 *rāśi* names in order. Use a phone-recording trick: record yourself saying them aloud, listen back twice a day. Test yourself by saying them without looking, in the order *Meṣa* → *Mīna*.
- Bonus: which *rāśi* is the Sun in right now? (Check the *nirayana* table; today’s date determines the answer.)

Differentiation

- **One tier up:** The *nirayana* table here is based on the Lahiri *ayanāṁśa* (~24° offset from the equinox). If you use the *sāyana* (tropical) system instead — used by Western horoscope columns — the Sun is in *Meṣa* / Aries from **21 March** to **20 April**, not 14 April to 14 May. Why the difference? Day 7 explains. (Hint: precession.)

- **One tier down:** Memorise just 4 names: *Meṣa* (Apr), *Karka* (Jul), *Tulā* (Oct), *Makara* (Jan). These are the four “anchors” — they fall near the solstices and equinoxes. The rest can come over the week.

Teacher notes

- Some students will get confused that their newspaper “Sun sign” doesn’t match the *nirayana* dates here. Reassure: this is expected — Western horoscope columns use the *sāyana* system; this table uses *nirayana*. They differ by ~24 days. Day 7 explains.
- The Sanskrit-Greek correspondence is genuinely striking and a good talking point. The historical reason: contact through the Hellenistic world after Alexander’s campaigns, with bidirectional flow.

Citation(s) used in this lesson

- **Sūrya-siddhānta** — paraphrase of the 12-*rāśi* division and the Sun’s mean angular speed. See [sources.md](#).